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Apparatus and method for producing an enriched medical suspension

Abstract

A medical fluid suspension generating delivery apparatus includes a dual syringe assembly including a first receptacle for a pressurized chemical solution and a second receptacle for a medical solution. The medical fluid suspension generating delivery apparatus also includes a Venturi-agitating tip assembly generating an enriched medical suspension from the pressurized chemical solution and a second receptacle for a medical solution.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) This invention relates to an apparatus for producing an enriched medical suspension.

Description of Related Art

(2) The present invention utilizes the Venturi effect to produce an enriched medical suspension for use in various applications. The apparatus of the present invention is simple to manufacture and use because it does not require an impeller and incorporated fan in order to create and dispense the enriched medical suspension.

(3) The Venturi effect is an example of Bernoulli's principle, in the case of incompressible fluid, flow through a tube or pipe with a constriction in it. The fluid velocity must increase through the constriction to satisfy the equation of continuity, while its pressure must decrease due to conservation of energy; the gain in kinetic energy is supplied by a drop in pressure or a pressure gradient force.

(4) The limiting case of the Venturi effect is choked flow, in which a constriction in a pipe or channel limits the total flow rate through the channel because the pressure cannot drop below zero in the constriction. Choked flow is used to control the delivery rate of water and other fluids through spigots and other types of valves. The portable apparatus of the present invention utilizes a source of pressurized medical fluid, to produce the desired pressure and flow for the effective creation of an enriched medical suspension.

SUMMARY

- (5) In one aspect a medical fluid suspension generating delivery apparatus includes a dual syringe assembly including a first receptacle for a pressurized chemical solution and a second receptacle for a medical solution. The medical fluid suspension generating delivery apparatus also includes a Venturi-agitating tip assembly generating an enriched medical suspension from the pressurized chemical solution and a second receptacle for a medical solution.
- (6) In some embodiments the dual syringe assembly includes a syringe body including a receptacle support having a first recess in which the first receptacle is mounted and a second recess in which the second receptacle is mounted.
- (7) In some embodiments sizes of the first receptacle and the second receptacle are different.
- (8) In some embodiments the first and second receptacles are oriented parallel to each other such that outlets of the respective first and second receptacles are aligned for connection with respective input cannulas of a dual lumen catheter connecting the pressurized chemical solution and the medical solution to the Venturi-agitating tip assembly.
- (9) In some embodiments the first receptacle includes a cylindrical body with a first end shaped and dimensioned to receive a first plunger and a second end defining the outlet of the first receptacle, and the second receptacle includes a cylindrical body with a first end shaped and dimensioned to receive a second plunger and a second end defining the outlet of the second receptacle.
- (10) In some embodiments an actuation member engages the first and second plungers such that the first and second plungers are moved in a coordinated manner.
- (11) In some embodiments the actuation member and the first and second plungers are oriented such that upon actuation a small amount of the pressurized chemical solution flows through the Venturi-agitating tip assembly before the medical solution flows therethrough.
- (12) In some embodiments the actuation member is fixedly attached to the free end of the first plunger.
- (13) In some embodiments a controlled actuation assembly is oriented in alignment with the free end of the second plunger.
- (14) In some embodiments a cup at the free end of the second plunger fits about the second end of the controlled actuation member. When the dual syringe assembly is filled and ready for use, the controlled actuation member is spaced from the free end of the second plunger, and when the thumb actuation member is first pressed downwardly only the first plunger is moved and the second plunger is not moved until the thumb actuation member has moved downwardly a distance sufficient to bring the second end of the controlled actuation member into contact with the free end of the second plunger.
- (15) Other objects and advantages of the present invention will become apparent from the following detailed description when viewed in conjunction with the accompanying drawings, which set forth certain embodiments of the invention.
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Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a perspective view of a dual syringe assembly prior to actuation.
- (2) FIG. 2 is a perspective view of the dual syringe assembly after use.
- (3) FIGS. 3, 4, and 5 are respectively a front elevation view, a top plan view, and a rear elevation view of the receptacle support.
- (4) FIG. 6 is a top plan view of the thumb actuation member.
- (5) FIG. 7 is a perspective view of the medical fluid suspension generating delivery apparatus.
- (6) FIGS. 8A, 8B, and 8C respectively show a longitudinal cross-sectional view, a perspective view and a perspective cross sectional view of a Venturi-agitating tip assembly in accordance with an alternate embodiment.

(7) FIGS. **9A**, **9B**, **9C**, and **9D** are respectively a perspective view, a longitudinal cross-sectional perspective view, an exploded view and a lateral cross-sectional view of a Venturi-agitating tip assembly in accordance with an alternate embodiment.

(8) FIGS. **10A**, **10B**, **10C**, **10D**, and **10E** are respectively a perspective view, an exploded view, a front partial cross-sectional view, a rear partial cross-sectional view and a lateral cross-sectional view of a Venturi-agitating tip assembly in accordance with an alternate embodiment.

(9) FIGS. **11A**, **11B**, **11C**, **11D**, **11E**, and **11F** are respectively a side elevation view, a detailed side elevation view, a detailed side elevation view, a perspective view, a longitudinal cross-sectional view, and a lateral cross-sectional view of a Venturi-agitating tip assembly in accordance with an alternate embodiment.

DETAILED DESCRIPTION

(10) The detailed embodiments of the present invention are disclosed herein. It should be understood, however, that the disclosed embodiments are merely exemplary of the invention, which may be embodied in various forms. Therefore, the details disclosed herein are not to be interpreted as limiting, but merely as a basis for teaching one skilled in the art how to make and/or use the invention.

(11) Referring to the various figures and embodiments, the medical fluid suspension generating delivery apparatus **100** for performing medical procedures includes a Venturi-agitating tip assembly (various embodiments of which are described below). Each of the Venturi-agitating tip assembly includes a multi-channel arrangement at a proximal first end thereof and a tip at a distal second end thereof. The delivery apparatus **100** also includes a dual syringe assembly **10** fluidly connected to the multi-channel arrangement at the proximal first end of the Venturi-agitating tip assembly. It is appreciated multiple embodiments of a Venturi-agitating tip assembly and the dual syringe assembly **10** is adapted work with any of the disclosed embodiments. In describing the dual syringe assembly **10** it is described with reference to the Venturi-agitating tip assembly **280** disclosed with reference to FIG. 7.

(12) The dual syringe assembly **10** includes a first receptacle **12** for a pressurized chemical solution **14** and a second receptacle **16** for the medical solution **18**. The pressurized chemical solution **14** and the medical solution **18** are combined within the Venturi-agitating tip assembly **280** in a manner generating an enriched medical suspension **20** that is ultimately dispensed from the suspension delivery apparatus **100**. The enriched medical suspension **20** is then sprayed from the Venturi-agitating tip assembly **280**. In accordance with a preferred embodiment, outlets **22** at the tip of the Venturi-agitating tip assembly **280** are oriented such that the enriched medical suspension **20** is sprayed at an angle of 25 degrees to 65 degrees, preferably 45 degrees, relative to the longitudinal axis of the Venturi-agitating tip assembly **280** to ensure the enriched medical suspension **20** is sprayed directly onto the inner wall of the lumen being treated and is not entrained within the fluid flowing in the lumen. This prevents the need for the application of the enriched medical suspension **20** down the middle of the lumen and waiting for the enriched medical suspension **20** to disperse out to the inner wall of the lumen. While the various embodiments disclosed below show the Venturi-agitating tip assembly with three (3) outlets, it is contemplated as many as 16 outlets may be positioned about the tip of the Venturi-agitating tip assembly for spraying the enriched medical suspension. Further still, it is appreciated the outlets may be covered with membranes enhancing delivery of the enriched medical suspension. A method in accordance with the apparatus is also disclosed.

(13) With reference to FIGS. **1** to **6**, the dual syringe assembly **10** is disclosed in detail. The dual syringe assembly **10** includes a syringe body **22** including a first receptacle **12** for a pressurized chemical solution **14** and a second receptacle **16** for the medical solution **18** for dispensing from outlets **24**, **26** of the dual syringe assembly **10**. Referring to FIGS. **1** to **5**, the syringe body **22** includes a receptacle support **50** having a first recess **52** in which the first receptacle **12** is mounted and a second recess **54** in which the second receptable **16** is mounted. In accordance with a

disclosed embodiment, the first and second receptacles **12**, **16** are secured to the first and second recesses **52**, **54** via adhesive, although it is appreciated the coupling could be achieved via a variety of known mechanisms.

(14) The dual syringe assembly **10** therefore includes a cylindrical first receptacle **12** and a cylindrical second receptacle **16**. The first and second receptacles **12**, **16** are oriented parallel to each other such that the outlets **24**, **26** of the respective first and second receptacles **12**, **16** are aligned for connection with respective input cannulas of the dual lumen catheter **260** connecting the pressurized chemical solution **14** and the medical solution **18** to the Venturi-agitating tip assembly **280**.

(15) The first receptacle **12** includes a cylindrical body **28** with a first end **28a** shaped and dimensioned to receive a first plunger **30** and a second end **28b** defining the outlet **24** of the first receptacle **12**. The second receptacle **16** includes a cylindrical body **32** with a first end **32a** shaped and dimensioned to receive a second plunger **34** and a second end **32b** defining the outlet **26** of the second receptacle **16**. As will be appreciated based upon the following disclosure relating to the operation of the delivery apparatus **100**, the sizes of the first receptacle **12** and the second receptacle **16** are different to accommodate desired quantities of the pressurized chemical solution **14** and the medical solution **18** for passage through the Venturi-agitating tip assembly **280** and dispensing as disclosed below. With this in mind, the first receptacle **12** is sized to accommodate volumes from 2 ml to 60 ml, and the second receptacle **16** is sized to accommodate volumes from 2 ml to 20 ml. Each of the first outlet **24** and the second outlet **26** includes a one-way valve **36**, **38** to ensure that the pressurized chemical solution **14** and the medical solution **18** only flows out of the dual syringe assembly **10**.

(16) Actuation of the first and second plungers **30**, **34** is controlled by a single thumb actuation member **40** that is fixedly linked to the free end **30f** of the first plunger **30** and contacts the second end **34f** of the second plunger **34** via a controlled actuation member **42**. The single thumb actuation member **40** engages the first and second plungers **30**, **34** such that the first and second plungers **30**, **34** are moved in a coordinated manner. Movement of the single thumb actuation member **40** is further controlled by a guide rod **43** that extends downwardly from the member thumb actuation member **40** and passes through an alignment recess **45** in the receptacle support **50**. As will be appreciated based upon the following disclosure, and in accordance with a disclosed embodiment, the thumb actuation member **40** and the first and second plungers **30**, **34** are oriented such that upon actuation a small amount of the pressurized chemical solution **14** flows through the Venturi-agitating tip assembly **280** before the medical solution **18** flows therethrough.

(17) As discussed above, the thumb actuation member **40** is fixedly attached to the free end **30f** of the first plunger **30**. The thumb actuation member **40** includes a plate member **44** having an upper surface **44u** shaped and dimensioned for engagement with the thumb of a user and a lower surface **44l** coupled to the free end **30f** of the first plunger **30**. The controlled actuation member **42** also extends from the lower surface **44l** of the plate member **44**.

(18) The controlled actuation member **42** is a cylindrical member with the first end **46** coupled to the lower surface **44l** of the plate member **44** and the second end **48** facing and in alignment with the free end **34f** of the second plunger **34** such that a cup **35** at the free end **34f** of the second plunger **34** fits about the second end **48** of the controlled actuation member **42**. However, and when the dual syringe assembly **10** is filled and ready for use, the controlled actuation member **42** is spaced from the free end **34f** of the second plunger **34**. As such, and when the thumb actuation member **40** is first pressed downwardly, only the first plunger **30** is moved. The second plunger **34** is not moved until the thumb actuation member **42** has moved downwardly a distance sufficient to bring the second end **48** of the controlled actuation member **42** into contact with the free end **34f** of the second plunger **34**. At this point both the first and second plungers **30**, **34** are moved forcing the pressurized chemical solution **14** and the medical solution **18** out of the dual syringe assembly **10**. Movement of the cup **35**, and ultimately the second plunger **34**, relative to the controlled actuation

member **42** is limited by a radially extending ridge **48r** at the second end **48** of the controlled actuation member **42** that interact with an inwardly extending ridge **35r** at the open end of the cup **35**.

(19) With reference to FIG. 7, the entire delivery apparatus **100** is shown. The suspension delivery catheter **2** includes a dual lumen catheter **260** connecting a Venturi-agitating tip assembly **280** to the pressurized chemical solution **14** and the medical solution **18** from the dual syringe assembly **10**. As mentioned above, the dual syringe assembly **10** includes one-way valves at its outlets to ensure only flows out of the dual syringe assembly **10**. The suspension delivery catheter **2** includes a first end (or distal end) **262** having the Venturi-agitating tip assembly **280** and a second end (or proximal end) **264** to which the dual syringe assembly **10** is fluidly connected for the passage of the pressurized chemical solution **14** and the medical solution **18**. As will be appreciated based upon the following disclosure, the dual lumen catheter **260** is connected to the Venturi-agitating tip assembly **280** by securing the Venturi-agitating tip assembly **280** to a first lumen **272** and a second lumen **274** of the dual lumen catheter **260**, respectively. The provision of the Venturi-agitating tip assembly **280** at the very end of the catheter allows for the mixing of the pressurized chemical solution **14** and the medical solution **18** immediately adjacent the discharge point.

(20) A micro-hose **256** connects the dual syringe assembly **10** to the first lumen **272** of the dual lumen catheter **260** at a proximal first end **266** of the dual lumen catheter **260** for the transmission of the pressurized chemical solution **14** from dual syringe assembly **10** to the Venturi-agitating tip assembly **280**. As such, pressurized chemical solution **14** leaving the dual syringe assembly **10** enters the first lumen **272** of the dual lumen catheter **260** via the micro-hose **256**. After passing through the first lumen **272** of the dual lumen catheter **260**, the pressurized medical chemical solution **14** enters the Venturi-agitating tip assembly **280** of the suspension delivery catheter **2**. As will be explained below in greater detail, the medical suspension **20** composed of the chemical solution **14** and the medical solution **18** generated at the Venturi-agitating tip assembly **280** is directly applied to a vein or artery requiring treatment with the medical suspension **20**.

(21) As to the connection of the medical solution **18** to the suspension delivery catheter **2**, the medical solution **18** is delivered to the second lumen **274** of the dual lumen catheter **260** at the proximal first end **266** thereof, and ultimately to the Venturi-agitating tip assembly **280**, via a container, in particular, the syringe **290**, connected to the second lumen **274** of the dual lumen catheter **260** by a supply line **216**. As mentioned above, the dual syringe assembly **10** includes one-way valves at its outlets to ensure that the pressurized chemical solution **14** and the medical solution **18** from the dual syringe assembly **10** only flow out of the dual syringe assembly **10**, preventing reflux back into the dual syringe assembly **10**. After passing through the second lumen **274** of the dual lumen catheter **260**, the medical solution **18** from the dual syringe assembly **10** travels into the Venturi-agitating tip assembly **280** where it is combined with pressurized chemical solution **14** from the dual syringe assembly **10** to form an enriched medical suspension **20**.

(22) The Venturi-agitating tip assembly **280** results in the spray of the enriched medical suspension **20** at an angle relative to a central axis of the Venturi-agitating tip assembly **280**. In accordance with a preferred embodiment, and as will be discussed below with regard to the various embodiments of the Venturi-agitating tip assembly **280**, the Venturi-agitating tip assembly **280** includes outlets **228** directing the enriched medical suspension **20** at an angle of 25 degrees to 65 degrees, preferably 45 degrees, relative to the central longitudinal axis of the Venturi-agitating tip assembly **280**. By orienting the outlets at an angle as disclosed herein, the enriched medical suspension **20** is directed toward the walls of the vessel in which it is being dispensed.

(23) It is appreciated various tip assemblies and enriched medical suspension generating structures may be employed in accordance with the present invention. In accordance with a first embodiment as shown with reference to FIGS. **8A-8C**, the Venturi-agitating tip assembly **380** employs a Venturi arrangement with a mixing chamber **324**. The Venturi-agitating tip assembly **380** has a proximal first end **380a** and a distal second end **380b**. The Venturi-agitating tip assembly **380** includes a

hollow cylindrical elongated body **310** having a proximal first end **312**, which coincides with the proximal first end **380a** of the Venturi-agitating tip assembly **380**, and a distal second end **314**. The proximal first end **380a** of the Venturi-agitating tip assembly **380** includes a multi-channel arrangement **381** including first and second inputs **316**, **318** for attachment to the dual lumen catheter **360**. The first and second inputs **316**, **318** respectively lead to a first channel **320** and a second channel **322** of the multi-channel arrangement **381** of the Venturi-agitating tip assembly **380**. The first and second channels **320**, **322** lead to, and are in fluid communication with, a mixing chamber **324** (which also forms part of the multi-channel arrangement **381**) located in the central portion **326** of the Venturi-agitating tip assembly **380**, that is, between the proximal first end **380a** and the distal second end **380b**. Located at the distal second end **380b** of the Venturi-agitating tip assembly **380**, and secured to the distal second end **314** of the elongated body **310**, is a spray tip **328** directing the enriched medical suspension in a spray pattern onto the inner lumen of a vessel. (24) The first channel **320** and the second channel **322** are interconnected in a manner creating a Venturi effect causing the pressurized chemical solution to effectively pull the medical solution through the second channel **322** and into the mixing chamber **324**. This is achieved by providing with the first channel **320** with a reduced diameter as it extends from the proximal first end **312** of the elongated body **310** (that is, the first end **320a** of the first channel **320**) to the central portion **326** of the Venturi-agitating tip assembly **380** (that is, the second end **320b** of the first channel **320**). In accordance with a preferred embodiment, the diameter of the first channel **320** decreases from a diameter of 0.038 inches adjacent the proximal first end **312** of the elongated body **310** to a diameter of 0.017 inches adjacent the mixing chamber **324**.

(25) As mentioned above, the second channel **322** is in fluid communication with the first channel **320**. This is achieved by the provision of a transverse channel **330** connecting the second end **320b** of the first channel **320** with the second end **322b** of the second channel **322**. In particular, the second channel **322** includes a first end **322a** adjacent the proximal first end **312** of the elongated body **310** and a second end **322b** adjacent the mixing chamber **324** (although not directly in fluid communication with the mixing chamber **324**) and the transverse channel **330**. In accordance with a preferred embodiment, the diameter of the second channel **322** is 0.031 inches and remains consistent as it extends from the first end **322a** thereof to the second end **322b** thereof.

(26) The first lumen **372** of the dual lumen catheter **360** supplies the pressurized chemical solution and the second lumen **374** supplies the medical solution. It is, however, appreciated the functions of the first and second lumens could be flipped in this embodiment as well as other embodiments. As such, the first lumen **372** is connected to, and in fluid communication with, the first channel **320** of the Venturi-agitating tip assembly **380** and the second lumen **374** is connected to, and in fluid communication with, the second channel **322** of the Venturi-agitating tip assembly **380**. In practice, the medical solution from the dual syringe assembly **10** travels through the second lumen **374** of the dual lumen catheter **360** and into the second channel **322** when pressurized chemical solution enters the first channel **320** and passes the transverse channel **330** as it flows into the mixing chamber **324** after being actuated and released from the dual syringe assembly **10**. The pressurized chemical solution entering, and passing through, the Venturi-agitating tip assembly **380** imparts negative pressure on the medical solution in the dual syringe assembly **10** and draws the medical solution from the dual syringe assembly **10** through the second channel **322**, through the second lumen **374** of the dual lumen catheter **360**, through the transverse channel **330**, and into the mixing chamber **324** due to the Venturi effect. The medical solution and the pressurized chemical solution are then mixed within the mixing chamber **324** to form an enriched medical suspension. The second plunger **30** is used to regulate or stop flow of medical solution or chemical solution from the dual syringe assembly **10**.

(27) The pressurized chemical solution and medical solution mixing in the mixing chamber **324** are then forced through the spray tip **328** from which the enriched medical suspension is sprayed upon the inner lumen of a vessel. The spray tip **328** includes a plurality of outlets **328a** oriented at an

angle of 25 degrees to 65 degrees, preferably 45 degrees, relative to the central longitudinal axis of the Venturi-agitating tip assembly **380**. The force of the pressurized chemical solution traveling through the Venturi-agitating tip assembly **380** and exiting through the spray tip **328** as part of an enriched medical suspension projects the enriched medical suspension from the distal second end **384** of the Venturi-agitating tip assembly **380** as a spray and onto the inner lumen of a vessel.

(28) In accordance with a second embodiment as shown with reference to FIGS. **9A** to **9D**, a Venturi-agitating tip assembly **480** employs a spray tip **428** in conjunction with a multi-channel arrangement **481** where the pressurized chemical solution and medical solution are mixed and forced through the spray tip **428**. The Venturi-agitating tip assembly **480** includes a proximal first end **480a** and a distal second end **480b**. The Venturi-agitating tip assembly **480** includes a hollow cylindrical elongated body **410** having a proximal first end **412**, which coincides with the proximal first end **480a** of the Venturi-agitating tip assembly **480**, and a distal second end **414**. The Venturi-agitating tip assembly **480** is adapted for use with a dual lumen catheter **460**, in particular a dual lumen catheter having concentric lumens, wherein the outer first lumen **472** is annular shaped for the passage of pressurized chemical solution (and has an outer diameter of 0.092 inches at the outer wall thereof and an inner diameter of 0.042 inches at the inner wall thereof) and the inner second lumen **474** is circular shaped for the passage of the medical solution (and has a diameter of 0.030 inches). The inner second lumen **474** is supported within the outer first lumen **472** by first and second radially extending rib members **473a**, **473b** (each having a thickness of 0.006 inches) that extend from the outer surface of the second lumen **474** to the inner surface of the outer first lumen **472**. In this way the outer first lumen **472** is divided into first and second semicircular passageways **475a**, **475b**.

(29) The proximal first end **480a** of the Venturi-agitating tip assembly **480**, in particular, the proximal first end **412** of the elongated body **410** is formed with two projections **432**, **434** shaped and dimensioned for engagement within the outer first lumen **472** of the catheter **460** in a manner blocking a substantial portion of the outer first lumen **472**. The two projections **432**, **434** are arcuate members shaped and dimensioned to respectively block substantial portions of the first and second semicircular passageways **475a**, **475b** while creating four small passageways **436**, each of approximately 0.031 inches (along the Y-axis as shown in FIG. **9D**) by 0.050 inches (along the X-axis as shown in FIG. **9D**) for the passage of pressurized chemical solution therethrough. The four small passageways **436** are defined by spaces existing between the edges of the arcuate members **432**, **434** and the first and second radially extending rib members **473a**, **473b**.

(30) The remainder of the Venturi-agitating tip assembly **480** includes a central mixing chamber **424** that is in fluid communication with the second lumen **474** and the four small passageways **436** feeding pressurized chemical solution from the first lumen **472**. Secured to, and closing off, the second end **414** of the elongated body **410** is a spray tip **428**, which is thereby at the distal second end **480b** of the Venturi-agitating tip assembly **480**. Attachment of the spray tip **428** to the elongated body **410** is achieved by providing the spray tip **428** with a projection **438** that seats within the opening at the second end **414** of the elongated body **410**.

(31) The first lumen **472** and the second lumen **474** are interconnected in a manner causing the pressurized chemical solution to effectively pull the medical solution through the second lumen **474** and into the mixing chamber **424**. In practice, the medical solution from the dual syringe assembly **10** travels through the second lumen **474** of the dual lumen catheter **460** and into the mixing chamber **424** when pressurized chemical solution passes through the four small passageways **436** and enters the mixing chamber **424** (where the medical solution from the syringe **290** and the pressurized chemical solution mix to form an enriched medical suspension) after being actuated and released from dual syringe assembly **10**. The pressurized chemical solution entering, and passing through, the mixing chamber **424** imparts negative pressure on the medical solution in dual syringe assembly **10** and draws the medical solution from the dual syringe assembly **10** through the second lumen **474** and into the mixing chamber **424**. The second plunger **30** is used to

regulate or stop flow of the medical solution from the dual syringe assembly **10**.

(32) The pressurized chemical solution and medical solution mixing in the mixing chamber **424** are then forced through the spray tip **428** from which an enriched medical suspension is sprayed upon the inner surface of a lumen. The spray tip **428** includes a plurality of outlets **428a** oriented at an angle of 25 degrees to 65 degrees, preferably 45 degrees, relative to the central longitudinal axis of the Venturi-agitating tip assembly **480**. The force of the pressurized chemical solution traveling through the Venturi-agitating tip assembly **480** and exiting through the spray tip **428** as part of an enriched medical suspension projects the enriched medical suspension from the distal second end **484** of the Venturi-agitating tip assembly **480** as a spray and onto the inner lumen of a vessel.

(33) In accordance with a third embodiment as shown with reference to FIGS. **10A** to **10E**, a Venturi-agitating tip assembly **780** employs a tip **728** in conjunction with a multi-channel arrangement **781** where the pressurized chemical solution and medical solution are mixed to form an enriched medical suspension and forced through the tip **728**. The Venturi-agitating tip assembly **780** includes proximal first end **780a** and a distal second end **780b**. The Venturi-agitating tip assembly **780** includes a hollow cylindrical elongated body **710** having a proximal first end **712**, which coincides with the proximal first end **780a** of the Venturi-agitating tip assembly **780**, and a distal second end **714**. The Venturi-agitating tip assembly **780** is adapted for use with a multi-lumen catheter **760**, in particular a triple lumen catheter having parallel lumens, wherein the first and second lumens **772**, **773** are circular shaped (each with a diameter of 0.039 inches) and are dimensioned for the passage of pressurized chemical solution and the third lumen **774** is semi-circular shaped (with a radius of 0.047 inches) and is dimensioned for the passage of the medical solution.

(34) The proximal first end **712** of the elongated body **710** at the proximal first end **780a** of the Venturi-agitating tip assembly **780** includes first, second and third inputs **716**, **717**, **718** for attachment to the multi-lumen catheter **760**. The first and second inputs **716**, **717** lead to a first channel **720** and the third input **718** to a second channel **722**. As such, the proximal first end **712** of the elongated body **710** at the proximal first end **780a** of the Venturi-agitating tip assembly **780** is formed with two circular tubular projections **732**, **734**, defining the first and second inputs **716**, **717**. The circular tubular projections **732**, **734** (each with an inner diameter of 0.027 inches and an outer diameter of 0.039 inches) are shaped and dimensioned for engagement within the first and second lumens **772**, **773** of the catheter **760** in a manner allowing for the flow of fluid from the first and second lumens **772**, **773** and into the Venturi-agitating tip assembly **780**. The two circular tubular projections **732**, **734** are shaped and dimensioned to fit within the first and second lumens **772**, **773** while maintaining passageways for the passage of pressurized chemical solution therethrough.

(35) The first and second channels **720**, **722** lead to, and are in fluid communication with, a mixing chamber **724** located in the central portion **726** of the Venturi-agitating tip assembly **780**, that is, between the proximal first end **712** and the distal second end **714** of the elongated body. Secured to the distal second end **714** of the elongated body **710**, and positioned at the distal second end **780b** of the Venturi-agitating tip assembly **780**, is a tip **728** having three outlets **728a**, **728b**, **728c** extending from the mixing chamber **724** to the exterior at the distal end of the Venturi-agitating tip assembly **780** at an angle of 25 degrees to 65 degrees, preferably 45 degrees, relative to the central longitudinal axis of the Venturi-agitating tip assembly **780**.

(36) The first channel **720** and the second channel **722** are interconnected in a manner creating a Venturi effect causing the pressurized chemical solution to effectively pull the medical solution through the second channel **722** and into the mixing chamber **724**. This is achieved by providing the first channel **720** with a reduced diameter (decreasing from 0.038 inches to 0.017 inches) as it extends from the proximal first end **712** of the elongated body **710** (that is, the first end **720a** of the first channel **720**) to the central portion **726** of the Venturi-agitating tip assembly **780** (that is, the second end **720b** of the first channel **720**). In accordance with a preferred embodiment, the

diameter of the first channel **720** decreases from a diameter of 0.038 inches adjacent the proximal first end **712** of the elongated body **710** to a diameter of 0.017 inches adjacent the mixing chamber **724**.

(37) As mentioned above, the second channel **722** is in fluid communication with the first channel **720**. This is achieved by the provision of a transverse channel **730** connecting the second end **720b** of the first channel **720** with the second end **722b** of the second channel **722**. In particular, the second channel **722** includes a first end **722a** adjacent the proximal first end **712** of the elongated body **710** and a second end **722b** adjacent the mixing chamber **724** (although not directly in fluid communication with the mixing chamber **724**) and the transverse channel **730**. In accordance with a preferred embodiment, the diameter of the second channel **722** is 0.047 inches and remains consistent as it extends from the first end **722a** thereof to the second end **722b** thereof.

(38) The first and second lumens **772**, **773** supply the pressurized chemical solution and the third lumen **774** supplies the medical solution. As such, the first and second lumens **772**, **773** are connected to, and in fluid communication with, the first channel **720** of the Venturi-agitating tip assembly **780**. The third lumen **774** is connected to, and in fluid communication with, the second channel **722** of the Venturi-agitating tip assembly **780**. In practice, the medical solution from dual syringe assembly **10** travels through third lumen **774** of multi-lumen lumen catheter **760** and into the second channel **722** when pressurized chemical solution enters the first channel **720** and passes the transverse channel **730** (having a size of 0.020 inches) into the mixing chamber **724** after being actuated and released from dual syringe assembly **10**. The pressurized chemical solution entering, and passing through, the Venturi-agitating tip assembly **780** imparts negative pressure on the medical solution in dual syringe assembly **10** and draws the medical solution from the dual syringe assembly **10** through second channel **722**, through the third lumen **774** of the dual lumen catheter **760**, and into the mixing chamber **724** due to the Venturi effect. The second plunger **30** is used to regulate or stop flow of medical solution from the dual syringe assembly **10**.

(39) The pressurized chemical solution and medical solution mixing in the mixing chamber **724** form an enriched medical suspension that is then forced through the outlets **728a-c** of the spray tip **728**. The force of the pressurized medical chemical solution traveling through the Venturi-agitating tip assembly **780** and exiting through the spray tip **728** as part of an enriched medical suspension projects the enriched medical suspension from the distal second end **784** of the Venturi-agitating tip assembly **780** as a spray and onto the inner lumen of a vessel.

(40) While the outlets of the spray tip in the embodiments disclosed above with reference to FIGS. **8A-C**, FIGS. **9A-D**, and **10A-E** are positioned along a forward portion of the spray tip, it is appreciated the outlets of the spray tip could be positioned along the outer circumferential wall of the spray tip as shown in FIG. **7**. Through the utilization of either position for the outlets, the enriched medical suspension is sprayed from the Venturi-agitating tip assembly onto the inner lumen of a vessel.

(41) For example, and with reference to FIG. **11A-11F**, a fourth embodiment with outlets **928a** of the spray tip **928** positioned along the outer circumferential wall **929** of the spray tip **928**. Those portions of the Venturi-agitating tip assembly **980** proximal to the spray tip **928**, for example, the mixing chamber **924**, the first channel **920**, and the second channel **922** are the same as found in the embodiment disclosed with reference to FIGS. **8A** to **8C**. However, the spray tip **928** includes a plurality of circumferentially oriented outlets **928a** positioned so as to dispense the enriched medical suspension through the outer circumferential wall **929** of the spray tip **928**.

(42) In accordance with the various embodiments described above, the enriched medical suspension exiting the Venturi-agitating tip assembly is directed to a vessel requiring treatment. In accordance with a preferred embodiment, the method for treatment in accordance with the present invention is achieved in the following manner. The first end of the suspension delivery catheter, that is, Venturi-agitating tip assembly is introduced into a diseased/varicose vein requiring treatment such that the first end of Venturi-agitating tip assembly is positioned beyond the section

of vein requiring treatment. The second end of the delivery catheter is coupled to the compressed medical fluid unit and the syringe. At this point, the compressed medical fluid unit is actuated to supply pressurized chemical solution to the suspension delivery catheter and an enriched medical suspension is produced at the Venturi-agitating tip assembly of the suspension delivery catheter. The enriched medical suspension sprays from the first end of Venturi-agitating tip assembly into the section of vein, artery, or other vessel requiring treatment. For example, and where the enriched medical suspension includes a sclerosant for the destruction of a diseased vein, as the catheter is withdrawn from the vein, the enriched medical suspension is sprayed into the vein at various segments causing the vein to go into spasm resulting in eventual destruction of the diseased vein. Where the present invention is used in the treatment of the arterial or venous system without the goal of spasm or vessel destruction, the enriched medical suspension is sprayed, or otherwise delivered, as required for the procedure being performed.

(43) It is appreciated this procedure can be performed under ultrasound guidance or radiograph in order for the physician to control the amount of liquid to mix with the pressurized chemical solution to form the enriched medical suspension.

(44) In addition to the treatment of diseased veins as discussed above, the present suspension delivery catheter may be used in the treatment of various vascular ailments. The potential treatments that may employ the present suspension delivery catheter include, but are not limited to the following, oncology medical solutions, microbeads, magnetic beads, or particles for thrombus treatment, metallic beads, or particles for thrombus treatment, embolics, driving drugs through the blood-brain barrier for neurological conditions, driving or delivering TPA (Tissue Plasminogen Activator) for thrombolytic usage, etc.

(45) It is appreciated that where microparticles are used in conjunction with the enriched medical suspension composed of the chemical and medical solutions, saline may be used with the microparticles, so as to place the microparticles into suspension.

(46) While this detailed description has set forth particularly preferred embodiments of the apparatus of this invention, numerous modifications and variations of the structure of this invention, all within the scope of the invention, will readily occur to those skilled in the art. Accordingly, it is understood that this description is illustrative only of the principles of the invention and is not limitative thereof.

(47) Although specific features of the invention are shown in some of the drawings and not others, this is for convenience only, as each feature may be combined with any and all of the other features in accordance with this invention.

(48) While the preferred embodiments have been shown and described, it will be understood that there is no intent to limit the invention by such disclosure, but rather, it is intended to cover all modifications and alternative constructions falling within the spirit and scope of the invention.

Claims

1. A medical fluid suspension generating delivery apparatus, comprising: a dual syringe assembly including a first receptacle for a pressurized chemical solution and a second receptacle for a medical solution, the first receptacle includes a cylindrical body with a first end shaped and dimensioned to receive a first plunger and a second end defining an outlet of the first receptacle, and the second receptacle includes a cylindrical body with a first end shaped and dimensioned to receive a second plunger and a second end defining an outlet of the second receptacle, the dual syringe assembly further including an actuation member engaging the first and second plungers such that the first and second plungers are moved in a coordinated manner, the actuation member comprising a controlled actuation assembly oriented in alignment with a free end of the second plunger; and a Venturi-agitating tip assembly generating an enriched medical suspension from the pressurized chemical solution and the medical solution; wherein when the dual syringe assembly is

filled and ready for use, the controlled actuation assembly is spaced from the free end of the second plunger, and when the actuation member is first pressed downwardly only the first plunger is moved and the second plunger is not moved until the actuation member has moved downwardly a distance sufficient to bring a second end of the controlled actuation assembly into contact with the free end of the second plunger.

2. The medical fluid suspension generating delivery apparatus according to claim 1, wherein the dual syringe assembly includes a syringe body including a receptacle support having a first recess in which the first receptacle is mounted and a second recess in which the second receptacle is mounted.

3. The medical fluid suspension generating delivery apparatus according to claim 1, wherein sizes of the first receptacle and the second receptacle are different.

4. The medical fluid suspension generating delivery apparatus according to claim 1, wherein the first and second receptacles are oriented parallel to each other such that the outlets of the respective first and second receptacles are aligned for connection with respective input cannulas of a dual lumen catheter connecting the pressurized chemical solution and the medical solution to the Venturi-agitating tip assembly.

5. The medical fluid suspension generating delivery apparatus according to claim 1, wherein the actuation member and the first and second plungers are oriented such that upon actuation a small amount the pressurized chemical solution flows through the Venturi-agitating tip assembly before the medical solution flows therethrough.

6. The medical fluid suspension generating delivery apparatus according to claim 5, wherein the actuation member is fixedly attached to a free end of the first plunger.

7. The medical fluid suspension generating delivery apparatus according to claim 1, wherein a cup at the free end of the second plunger fits about the second end of the controlled actuation assembly.

8. A medical fluid suspension generating delivery apparatus, comprising: a dual syringe assembly comprising: a first receptacle for a pressurized chemical solution and a second receptacle for a medical solution, the first receptacle includes a cylindrical body with a first end shaped and dimensioned to receive a first plunger and a second end defining an outlet of the first receptacle, and the second receptacle includes a cylindrical body with a first end shaped and dimensioned to receive a second plunger and a second end defining an outlet of the second receptacle; the first and second receptacles are oriented parallel to each other such that the outlets of the respective first and second receptacles are aligned for connection with respective input cannulas of a dual lumen catheter connecting the pressurized chemical solution and the medical solution to a Venturi-agitating tip assembly; the dual syringe assembly further including an actuation member engages the first and second plungers such that the first and second plungers are moved in a coordinated manner, the actuation member is fixedly attached to a free end of the first plunger and the actuation member includes a controlled actuation assembly is oriented in alignment with a free end of the second plunger; the Venturi-agitating tip assembly generating an enriched medical suspension from the pressurized chemical solution and the medical solution; wherein the actuation member and the first and second plungers are oriented such that upon actuation a small amount the pressurized chemical solution flows through the Venturi-agitating tip assembly before the medical solution flows therethrough and a cup at the free end of the second plunger fits about a second end of the controlled actuation assembly, wherein when the dual syringe assembly is filled and ready for use, the controlled actuation assembly is spaced from the free end of the second plunger, and when the actuation member is first pressed downwardly only the first plunger is moved and the second plunger is not moved until the actuation member has moved downwardly a distance sufficient to bring the second end of the controlled actuation assembly into contact with the free end of the second plunger.
